

Chapter 2: Probability

1) $P(A)$: Xác suất của biến cố A

$$P(A) = \frac{|A|}{|\Omega|}$$

$$0 \leq P(A) \leq 1$$

(*) Some probability formulas

1) Addition rule - (OR) : to find $P(A \cup B)$

a) If a and b are ~~mut~~ mutually events

$$P(A \cup B) = P(A) + P(B)$$

$$\rightarrow P(E_1 \cup \dots \cup E_n) = P(E_1) + P(E_2) + \dots + P(E_n)$$

b) If A and B are any events

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

2) Multiplication rule - (AND) : to find $P(A \cap B)$

$$P(A \cap B) = P(A) \cdot P(B|A)$$

$$= P(B) \cdot P(A|B)$$

A and B is independent

$$\Rightarrow P(A \cap B) = P(A) \cdot P(B)$$

37 Conditional probability

(Cho sẵn giá trị $P(A \cap B)$)

$$P(A|B) = P(A \mid \text{given that condition B})$$

[if

$$= \frac{P(A \cap B)}{P(B)}$$

$$P(B|A) = \frac{P(A \cap B)}{P(A)}$$

47 Complement rule

$$P(\bar{A}) = 1 - P(A)$$

⊛ Total probability

$$P(B) = P(B \cap E_1) + P(B \cap E_2) + \dots + P(B \cap E_n)$$

$$= P(B|E_1) \cdot P(E_1) + P(B|E_2) \cdot P(E_2) + \dots + P(B|E_n) \cdot P(E_n)$$

⊛ Bayes' Theorem

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$

CHAPTER 3

Chapter 3: Discrete Random Variables

3.17 Discrete Random Variables

3.27

⊛ Probability mass function
(mật độ xác suất)

$$X = \{x_1, x_2, x_3, \dots, x_n\}$$

$$\rightarrow f(x_i) = P(X = x_i)$$

$$\rightarrow f(x_i) \geq 0 \text{ và } \sum_{i=1}^n f(x_i) = 1$$

$$\rightarrow P(a \leq X \leq b) = \sum_{a \leq x_i \leq b} f(x_i)$$

3.37 Cumulative distribution functions

$$F(x) = P(X \leq x) = \sum_{x_i \leq x} f(x_i)$$

$$\rightarrow 0 \leq F(x) \leq 1$$

$$2) \text{ If } x \leq y \text{ then } f(x) \leq f(y)$$

$$\Delta P(a \leq x \leq b) = P(x \leq b) - P(x < a)$$

3.47 Mean and variance

$$\rightarrow \mu = E(X) = \sum x_i f(x_i)$$

$$\rightarrow \sigma^2 = V(X) = E(X - \mu)^2 = \sum x_i^2 f(x_i) - \mu^2$$

- standard deviation of X is
(độ lệch chuẩn $\sigma = \sqrt{V(x)}$)

3.57 Discrete uniform distribution

Def: $f(x_i) = \frac{1}{n}$.

→ sx mọi $\sigma^2 = \text{nhau}$.

⇒ Mean and Variance

$$+ E(X) = \frac{a+b}{2} = \mu$$

$$+ \sigma^2 = V(x) = \frac{(b-a+1)^2 - 1}{12}$$

3.67 Binomial distribution

$$f(x) = C_x^n p^x (1-p)^{n-x}$$

$$x = 0, 1, \dots, n.$$

$$+ \text{Mean } \mu = E(X) = np$$

$$+ \text{Variance } \sigma^2 = V(x) = np(1-p)$$

3.77 Geometric and negative binomial distribution

⊛ Geometric distribution (p^2 hình học)

$$f(x) = (1-p)^{x-1} p \quad x=1, \dots$$

(xác suất của phép thử cho đến khi có 1 lần thử thành công)

Mean $\mu = E(x) = \frac{1}{p}$

$$\sigma^2 = V(x) = \frac{(1-p)}{p^2}$$

⊛ Negative Binomial distribution (p^2 nhị thức âm)

Số xuất số lần thử cho đến khi thành công
 r lần

$$f(x) = \binom{x-1}{r-1} (1-p)^{x-r} p^r$$

Mean $\mu = \frac{r}{p}$

$$\sigma^2 = E(x) = \frac{r(1-p)}{p^2}$$

3.87 Hyper-geometric distribution

$$f(x) = \frac{C_x^k \cdot C_{n-x}^{n-k}}{C_N^N}$$

+1 Mean: $\mu = E(x) = np$

+2 Variance

$$\sigma^2 = V(x) = np(1-p) \cdot \frac{N-n}{n-1} \quad p = \frac{k}{N}$$

3.97 Poisson distribution

X : number of events that can appear in a fixed interval with parameter $\lambda > 0$

λ : the average number of events that in the interval

$$f(x, \lambda) = \frac{e^{-\lambda} \cdot \lambda^x}{x!} \text{ for } x = 0, 1, \dots, \infty$$

$$\mu = \sigma^2 = \lambda$$

CHAPTER 4

Chapter 4 : Continuous Random Variable

1) Continuous Random Variable

+> Probability density function (pdf) of a continuous random variable X is a function

$$+> f(x) \geq 0$$

$$+> \int_{-\infty}^{+\infty} f(x) dx = 1$$

$$+> P(a \leq x \leq b) = \int_a^b f(x) dx$$

$$P(x_1 \leq X \leq x_2) = \begin{cases} P(x_1 \leq x < x_2) \\ P(x_1 < x \leq x_2) \\ P(x_1 \leq x < x_2) \end{cases}$$

In continuous random variable

$$P(X = x_i) = 0$$

2) Cumulative distribution function

$$F(x) = \int_{-\infty}^x f(x) dx$$

$$\text{for } -\infty < x < +\infty$$

$$f(x) = F'(x)$$

$$P(a < X < b) = F(b) - F(a)$$

$$\text{Mean } \mu = E(X) = \int_{-\infty}^{+\infty} x f(x) dx.$$

$$\begin{aligned} \sigma^2 = V(X) &= \int_{-\infty}^{+\infty} (x - \mu)^2 f(x) dx \\ &= \int_{-\infty}^{+\infty} x^2 f(x) dx - \mu^2 \end{aligned}$$

$$\rightarrow \text{standard deviation } \sigma = \sqrt{V(X)}$$

37 Continuous uniform distribution

$$\text{pdf: } f(x) = \begin{cases} \frac{1}{b-a} & a \leq x \leq b \\ 0 & \text{otherwise} \end{cases}$$

$$\text{Mean } \mu = E(X) = \frac{b+a}{2}$$

$$\sigma^2 = V(X) = \frac{(b-a)^2}{12}$$

$$\sigma = \sqrt{V(X)}$$

$$\text{cdf: } F(x) = \begin{cases} 0 & x < a \\ \frac{x-a}{b-a} & a \leq x < b \\ 1 & b \leq x \end{cases}$$

27 AA Normal distribution ($X \sim N(\mu, \sigma^2)$)

$$f(x) = \begin{cases} f(x) > 0 \\ \text{bell-shaped} \\ \text{symmetric} \end{cases}$$

$$f(x) = \frac{1}{\sqrt{2\pi} \sigma} \cdot e^{-\frac{1}{2} \left(\frac{x-\mu}{\sigma} \right)^2}$$

② Chuyển sang $z \sim N(0, 1)$

$$1) P(X < a) = P\left(z < \frac{a - \mu}{\sigma}\right)$$

$$2) P(X > a) = 1 - P(X < a) = 1 - P\left(z \leq \frac{a - \mu}{\sigma}\right)$$

$$3) P(b < X < a) = P(X < b) - P(X < a) \\ = P\left(z < \frac{b - \mu}{\sigma}\right) - P\left(z < \frac{a - \mu}{\sigma}\right)$$

③ Normal approximation to the binomial

a) Normal approximation to binomial

$$\mu = np \\ \sigma^2 = np(1-p)$$

$$1) P(X_{bi} \leq a) = P(X_{norm} \leq a + 0.05) \\ = P\left(z < \frac{a + 0.05 - \mu}{\sigma}\right)$$

$$2) P(X_{bi} > a) = P(X_{norm} > a - 0.05) \\ = 1 - P(X_{norm} \leq a - 0.05) \\ = 1 - P\left(z \leq \frac{a - 0.05 - \mu}{\sigma}\right)$$

④ Note: mình
viết nhầm
 $0.05 \rightarrow 0.5$

~~1) $P(X_{bi} \leq a)$~~

$$1) P(a < X_{bi} < b) = P(a - 0.5 \leq X_i \leq b + 0.5) \\ = P(X_{norm} \leq b + 0.5) - P(X_{norm} \leq a - 0.5)$$

b) Poisson distribution

$$X \sim P(\lambda), T=1$$

$$\rightarrow z = \frac{x - \lambda}{\sqrt{\lambda}}$$

$$+7 P(X_{\text{poisson}} \leq a) = P(X_{\text{norm}} \leq a + 0.5)$$

$$+7 P(a < X_{\text{poisson}})$$

$$= P(X_{\text{norm}} > a - 0.5)$$

$$= 1 - P\left(z \leq \frac{a - 0.5 - \lambda}{\sqrt{\lambda}}\right)$$

④ Exponential distribution.

$$P(X=x) = P(N=0)$$

$$= \frac{e^{-\lambda x}}{0!} (\lambda x)^0 = e^{-x \cdot \lambda}$$

$$P(X=x) = e^{-\lambda x}$$

$$\rightarrow P(X < x) = 1 - e^{-\lambda x}$$

$$f(x) = F(x) = \lambda e^{-\lambda x}$$

$$\text{Mean } E(X) = \frac{1}{\lambda}$$

$$\text{Variance } V(X) = \frac{1}{\lambda^2}$$

CHAPTER 7:

Chapter 7: Samp Sampling distribution

17 Point estimate for mean (μ)

$$\begin{cases} X \sim N(\mu, \sigma^2) \\ \text{sample } n \geq 30 \end{cases} \rightarrow \bar{X} \sim N(\mu, \sigma^2/n)$$

$$\begin{cases} \mu_{\bar{X}} = \mu \\ \sigma_{\bar{X}}^2 = \frac{\sigma^2}{n} \end{cases} \rightarrow \bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

$$\text{standard error } s: \frac{\sigma}{\sqrt{n}}$$

⑧ Point estimate for difference ~~mean~~ mean
($\bar{X} - \bar{Y}$)

$$X \begin{cases} X \sim N(\mu_1, \sigma_1^2) \\ n_1 \geq 30 \end{cases} \rightarrow \bar{X} \sim N\left(\mu_1, \frac{\sigma_1^2}{n_1}\right)$$

$$Y \begin{cases} Y \sim N(\mu_2, \sigma_2^2) \\ n_2 \geq 30 \end{cases} \rightarrow \bar{Y} \sim N\left(\mu_2, \frac{\sigma_2^2}{n_2}\right)$$

$$\rightarrow \bar{X} - \bar{Y} \sim N\left(\mu_1 - \mu_2, \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}\right)$$

CHAPTER 8

CHAPTER 8: Statistical Intervals.

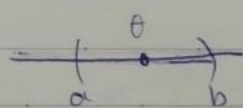
8.1. Confidence Interval on the mean of a normal distribution, variance known

$$\bar{X} \leadsto \mu \quad \left[\begin{array}{l} \text{normal} \\ n \gg 30 \end{array} \right] \rightarrow \bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

$$\hat{p} \sim p \quad (n \gg) \rightarrow \hat{p} \sim N\left(p, \frac{pq}{n}\right)$$

Ex: (a, b) $n = 36$, $\bar{x} = 66$ inches ($1 \text{ inch} = 2.54 \text{ cm}$)
 $\sigma = 3$ inches, $p = 95\%$

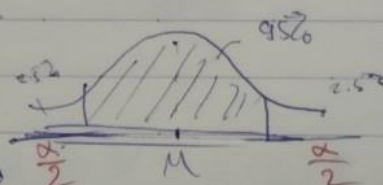
Find (a, b) confidence interval.



$$P(\mu - E \leq \bar{X} \leq \mu + E) = 95\%$$

$$\rightarrow P(\bar{X} \leq \mu - E) = 2.5\%$$

$$\Rightarrow P\left(Z < \frac{(\mu - E) - \mu}{\frac{\sigma}{\sqrt{n}}}\right) = 0.025$$



$$\Rightarrow E \approx 0.98 \Rightarrow (a, b)$$

with $100(1 - \alpha)\%$

$$\Rightarrow (a, b) = (65.02, 66.98)$$

Def: The confidence interval $\checkmark$ for parameter $\theta (\mu, p)$ is (l, u)

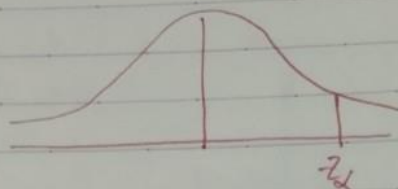
s.t.

$$P(l \leq \theta \leq u) = 1$$

lower upper

Error: $|\mu - \bar{x}| \leq E$
margin of error

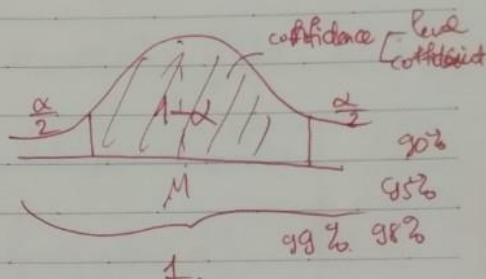
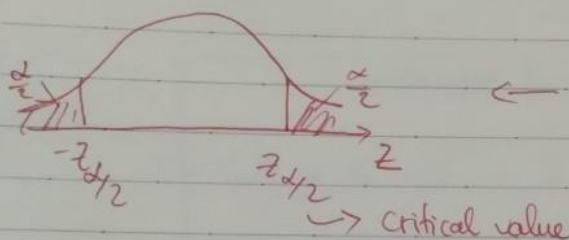
$$P(Z > z_\alpha) = \alpha$$



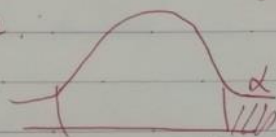
Critical value

(điểm tới hạn)

CI for mean (σ : known)
(confidence interval)



$$P(Z > z_\alpha) = \alpha$$



$z_\alpha \rightarrow$ critical value
(điểm phân vị)

$$P(Z > z_{\frac{\alpha}{2}}) = \frac{\alpha}{2}$$

$$\Leftrightarrow 1 - P(Z < z_{\alpha/2}) = \frac{\alpha}{2}$$

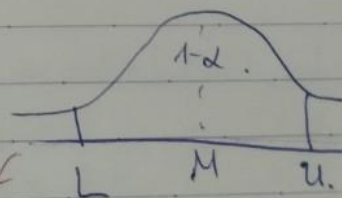
$$\Phi(z_{\alpha/2}) = 1 - \frac{\alpha}{2}$$

$$\Rightarrow z_{\alpha/2} = \text{NORMS.INV}(1 - \frac{\alpha}{2}) = \Phi(1 - \frac{\alpha}{2})$$

$1 - \alpha$	$z_{\alpha/2}$
90%	1.64
95%	1.96
98%	2.33
99%	2.58

$1 - \alpha$: confidence level

$$P(L \leq M \leq U) = 1 - \alpha$$



$$\bar{X} = N\left(\mu, \frac{\sigma^2}{n}\right) \Rightarrow Z = \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} \Rightarrow Z \sim N(0, 1)$$

$$P\left(\frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} \leq Z_{\alpha/2}\right) = 1 - \alpha$$

$\Downarrow$

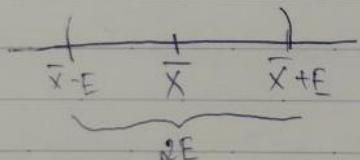
$$\Rightarrow Z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}} \leq \bar{X} - \mu \leq Z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}} \Leftrightarrow \bar{X} - Z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{X} + Z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$$

khoảng tin cậy cho Mean of population.

$$\bar{X} - Z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{X} + Z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$$

Ex. with $(1 - \alpha)$ 100% confidence level

$$E = Z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}} \Rightarrow CI = (\bar{X} \pm E)$$



Độ rộng của khoảng tin cậy.

$$P(\bar{X} - E \leq \mu \leq \bar{X} + E) = 1 - \alpha$$

$$\text{Width of CI} = 2E = 2 \cdot Z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$$

$$\Rightarrow |\mu - \bar{X}| \leq E \leftarrow \text{margin of error.}$$

Ex. CI for mean population

$$n = 36, \bar{X} = 66 \text{ inches (1 inches} = 2.54 \text{ cm)}$$

$$\sigma = 3 \text{ inches, } 95\%$$

Solution.

Check on distribution $\left\{ \begin{array}{l} \text{normal} \\ n > 30. \checkmark \end{array} \right.$

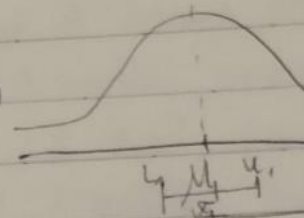
$$E = Z_{\frac{\alpha}{2}} \cdot \frac{\sigma}{\sqrt{n}} = 1.96 \cdot \frac{3}{\sqrt{36}} = 0.98$$

Sampling distribution

$$\Rightarrow CI = (\bar{X} \pm E)$$

$$\Rightarrow (66 - 0.98, 66 + 0.98)$$

$$\Rightarrow CI = (65.02, 66.98)$$



$\Rightarrow$ 95% confident that the true mean $\in (65.02, 66.98)$

$$E = z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$$

$$n = \left\lceil \frac{z_{\alpha/2}^2 \cdot \sigma^2}{E^2} \right\rceil$$

$$\Rightarrow E \downarrow \quad n \uparrow$$

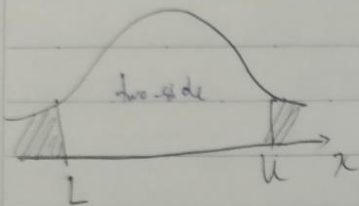
$$z_{\alpha/2} \downarrow \Rightarrow \frac{\alpha}{2} \uparrow \Rightarrow (1 - \alpha) \downarrow$$

Khi khoảng tin cậy giảm $\begin{cases} \rightarrow \text{độ tin cậy giảm} \\ \rightarrow n \text{ tăng} \end{cases}$

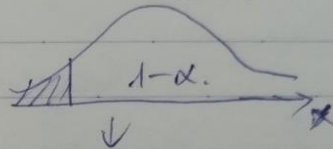
$$1 - \alpha = CI \neq 100\%$$

x: An economist is interested in studying the incomes of consumers in a particular region. The population standard deviation is known to be \$1000. A random sample of 64 individuals

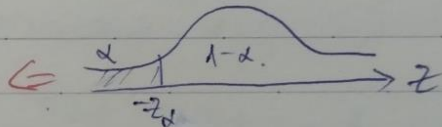
CI: Two-sided confidence bounds



One-side: $P(L \leq \bar{X}) = 1 - \alpha$



$$P\left(\frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} \geq -z_{\alpha}\right)$$



$$P\left(\bar{X} - \mu \geq -z_{\alpha} \cdot \frac{\sigma}{\sqrt{n}}\right) \Rightarrow P\left(\mu \leq \bar{X} - \underbrace{z_{\alpha} \cdot \frac{\sigma}{\sqrt{n}}}_E\right)$$

~~$\Rightarrow P(\mu \leq \bar{X})$~~

$(1 - \alpha) 100\%$ lower-confidence bound for mean is: $\bar{X} - E$

$(1 - \alpha) 100\%$ upper-confidence bound for mean is: $\bar{X} + E$

$$E = z_{\alpha} \cdot \frac{\sigma}{\sqrt{n}}$$

(2) CI for mean (σ unknown)

a) $n > 30$

($S \approx \sigma$)

standard deviation for sample

$$\Rightarrow CI = (\bar{X} \pm E)$$

$$\text{with } E = z_{\alpha/2} \frac{S}{\sqrt{n}}$$

Ex: $n = 50$

SD data

bài này $\rightarrow S, \bar{X}$

b) $n \leq 30$ (student distribution)

σ : unknown (không biết)

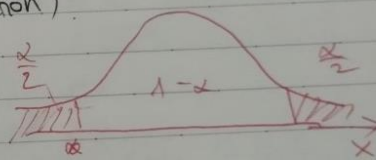
$S^2 \rightarrow \sigma^2$: unbiased

$$t = \frac{\bar{X} - \mu}{\frac{S}{\sqrt{n}}}$$

có thể bài này

$$t = \frac{S}{\sqrt{n}}$$

t - Student distribution



$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

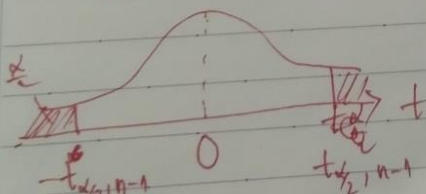
$$P(L \leq \bar{X} \leq U)$$

$$P(-z_{\alpha/2} \leq Z \leq z_{\alpha/2})$$

$$\frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

σ known

$$\frac{\bar{X} + \mu}{\frac{\sigma}{\sqrt{n}}}$$



Remark



t - distribution
student

- t - distribution : $\begin{cases} \text{bell-shaped} \\ \text{symmetric} \end{cases}$

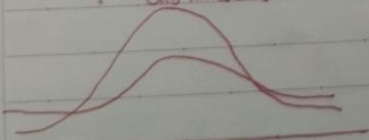
- mean = median = 0

- If $n \gg \infty \Rightarrow t_{\alpha/2, n-1} \approx z_{\alpha/2}$
($n > 150$)

Ex : $1 - \alpha = 90\%$, $n = 2001$.

$$t_{0.05, 2000} \approx z_{0.05} = 1.64$$

- t - distribution



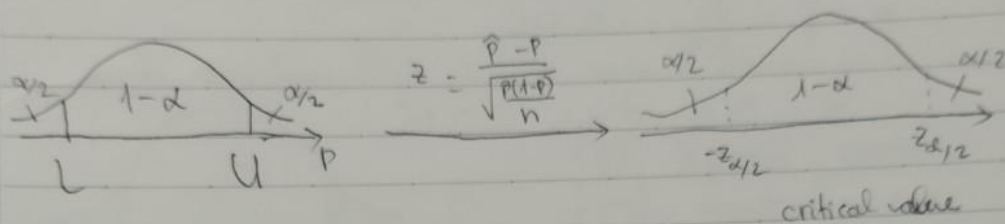
more area in tail than normal z

less area in center than normal z

XI LONG

③ CI for proportion (p) $\Rightarrow$ using z .

Recall Chap 7 $\hat{p} \sim p \rightarrow \hat{p} \sim N(p, \frac{p(1-p)}{n})$
 $\hat{p} = \frac{x}{n}$ mean variance



$$\Rightarrow P(L < p < U) = P(-z_{\alpha/2} < \frac{\hat{p} - p}{\sqrt{\frac{p(1-p)}{n}}} < z_{\alpha/2})$$

$$= P(\hat{p} - z_{\alpha/2} \sqrt{\frac{p(1-p)}{n}} < p < \hat{p} + z_{\alpha/2} \sqrt{\frac{p(1-p)}{n}})$$

① $(1-\alpha)$ 100% CI for proportion: $\hat{p} \pm E$

$$E = z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

$$(n \gg, \hat{p} \sim p \Rightarrow \hat{p} \approx p)$$

② Width of CI = $2E$

③ [Find sample size $\rightarrow$ Find $n = ?$
 How large a sample

$$n = \left\lceil \frac{z_{\alpha/2}^2 \cdot \hat{p}(1-\hat{p})}{E^2} \right\rceil$$

⚠ If $\hat{p}$ unknown $\Rightarrow \hat{p}(1-\hat{p})$ max.
 $\Leftrightarrow (\hat{p} - \hat{p}^2)' = 0$
 $\Leftrightarrow 1 - 2\hat{p} = 0$
 $\Rightarrow \hat{p} = 0.5$

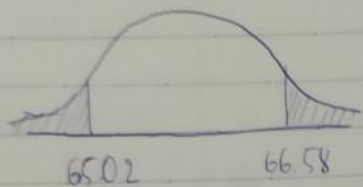
CHAPTER 9

CHAP 9: Test of hypothesis for a single sample (one).

< kiểm định giả thiết cho 1 sample >

Ex: 95% confidence interval for mean is (65.02, 66.58)

↓
95% confidence that the true mean is between 65.02 and 66.58



Critical region

Acceptance region.

• claim: $\mu = 60 \in \text{shaded box} \Rightarrow$ Reject claim

• claim: $\mu = 66 \in \text{white box} \Rightarrow$ Fail to reject claim

• claim: $\mu \neq 66 \Rightarrow \neg (\mu = 66) \Rightarrow$ Reject claim

$\rightarrow$ quá trình kiểm định.

Def:

① A statistical hypothesis is a statement about parameter of one or more population. (μ, σ^2, p)

Δ claim $\mu > 60$ ($\nexists \bar{x} > 66$)

claim $\sigma^2 = 2.05$ ($\nexists s^2 = 2.05$)

claim $p \leq 0.5$ ($\nexists \hat{p} \leq 0.5$)

② A procedure leading to a decision about a particular hypothesis is called "Test of hypothesis".

③ Claim $\Rightarrow$ $\left\{ \begin{array}{l} \text{null hypothesis: } H_0 \text{ (" = ")} \\ \text{alternative hypothesis: } H_1 \text{ (not " = ")} \end{array} \right.$

$$\begin{cases} H_0: \mu = 66 \\ H_1: \mu \neq 66 \end{cases}$$

b) claim $\mu > 66$

$$\begin{cases} H_0: \mu \leq 66 \text{ (or } \mu = 66) \\ H_1: \mu > 66 \end{cases}$$

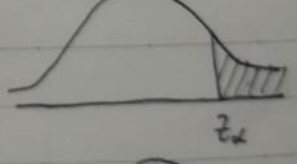
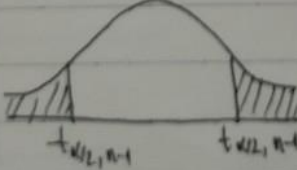
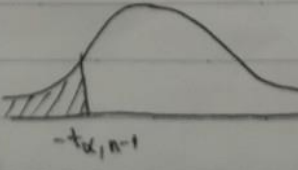
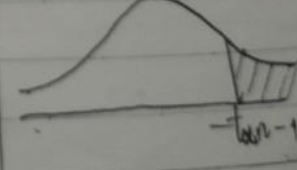
c) claim $p \geq 0.2$

$$\begin{cases} H_0: p \geq 0.2 \text{ (or } p = 0.2) \\ H_1: p < 0.2 \end{cases}$$

d) claim $p < 0.2$

$$\begin{cases} H_0: p \geq 0.2 \text{ (or } p = 0.2) \\ H_1: p < 0.2 \end{cases}$$

④ Find test statistic.

Claim.	$p = p_0$ $\mu = \mu_0$	$p \geq p_0$ $\mu \geq \mu_0$	$\mu > \mu_0$ $p > p_0$
$\begin{cases} H_0 \\ H_1 \end{cases}$	$\begin{cases} H_0: \mu = \mu_0 \\ H_1: \mu \neq \mu_0 \end{cases}$ (two-tail)	$\begin{cases} H_0: \mu \geq \mu_0 \\ H_1: \mu < \mu_0 \end{cases}$ (left tail)	$\begin{cases} H_0: \mu \leq \mu_0 \\ H_1: \mu > \mu_0 \end{cases}$ (right tail)
Critical value.			
$\begin{cases} z: \text{known} \\ (z_{\alpha/2} \text{ or } z_{\alpha}) \end{cases}$			
$\begin{cases} (t_{\alpha/2, n-1} \text{ or } t_{\alpha, n-1}) \\ z: \text{unknown} \end{cases}$			

I. ~~Scap~~ Sampling distribution

Find test statistic	
σ : known	$z_0 = \frac{\bar{X} - \mu_0}{\frac{\sigma}{\sqrt{n}}}$ $\begin{cases} z_0 \in \square \rightarrow \text{fail to reject } H_0 \\ z_0 \in \blacksquare \rightarrow \text{reject } H_0 \end{cases}$
σ : unknown	$t_0 = \frac{\bar{X} - \mu_0}{\frac{s}{\sqrt{n}}}$ $\begin{cases} t_0 \in \square \rightarrow \text{fail to reject } H_0 \\ t_0 \in \blacksquare \rightarrow \text{reject } H_0 \end{cases}$
for proportion (p)	$z_0 = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$ $\hat{p} = \frac{x}{n}$ $\begin{cases} z_0 \in \square \rightarrow \text{fail to reject } H_0 \\ z_0 \in \blacksquare \rightarrow \text{reject } H_0 \end{cases}$

Q. Enter 7: ~~Sample~~ Sampling distribution

	H_0 true	H_0 false
Reject H_0	Error type I = α ✓	✓ Error type I = β
Fail to reject H_0	✓	

$\alpha = P(\text{reject } H_0 | H_0 \text{ true})$: Significance level

$\beta = P(\text{fail to reject } H_0 | H_0 \text{ false})$: the power of test

Ex: 90% CI for mean is (65.02, 66.92), $\sigma = 20.5$, $n = 10$
Find probability of error type I?

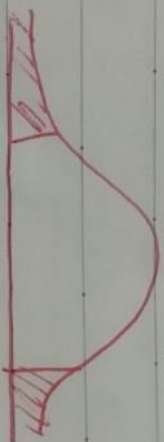
$$\begin{aligned}
 \Rightarrow \alpha &= P(\bar{X} < 65.02 | \mu = 66) + P(\bar{X} > 66.92 | \mu = 66) \\
 &= P\left(Z < \frac{65.02 - 66}{\frac{20.5}{\sqrt{100}}}\right) + 1 - P\left(Z < \frac{66.92 - 66}{\frac{20.5}{\sqrt{100}}}\right) \\
 &= 0.643
 \end{aligned}$$

* P-value (test of hypothesis)

① σ : known
for proportion.

$\Rightarrow$ using z .

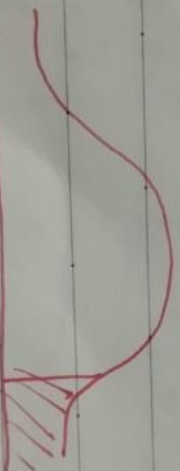
test statistics



$$P\text{-value} = P(Z > z_0)$$

$< \alpha \Rightarrow$ reject H_0
 $> \alpha \Rightarrow$ fail to reject

$$P\text{-value} = P(Z > z_0)$$



$$P\text{-value} = P(Z < z_0)$$

CHAPTER 10

(2) 0 : unknown $\Rightarrow$ using t
Same ①, but $z \rightarrow t$

Chapter 10 : Statistical inference for two samples

Recall.

Chap 7 :

$X \rightarrow$ sample n_1

[normal $\Rightarrow \bar{X} \sim N(\mu_1, \frac{\sigma_1^2}{n_1})$
 $n_1 \geq 30$

$Y \rightarrow$ sample n_2

[normal $\Rightarrow \bar{Y} \sim N(\mu_2, \frac{\sigma_2^2}{n_2})$
 $n_2 \geq 30$

$$\Rightarrow \bar{X} - \bar{Y} \sim N(\mu_1 - \mu_2, \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2})$$

Chap 8 :

$(1-\alpha)$ 100% CI for mean (μ) : $(\bar{X} \pm E)$
 $(1-\alpha)$ " " proportion (p) : $(\hat{p} \pm E)$

Chap 9 :

Test claim for mean (μ)
" " proportion (p)

① For difference mean ($\mu_1 - \mu_2$).

Point estimate for $\mu_1 - \mu_2$ $\bar{X}_1 - \bar{X}_2$	CI for $\mu_1 - \mu_2$ $(\bar{X}_1 - \bar{X}_2) \pm E$	Test hypothesis: $H_0: \mu_1 - \mu_2 = \Delta_0$
a) σ_1, σ_2 : known $\rightarrow$ using z .	$E = z_{\alpha/2} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$	Step 1, 2, 3 same table chap 9. Step 4 : Test statistic $z_0 = \frac{\bar{X}_1 - \bar{X}_2 - \Delta_0}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$

σ_1, σ_2 : unknown. $\rightarrow$ using t. i) assume $\sigma_1^2 = \sigma_2^2$ $(1 < \frac{s_{\text{large}}^2}{s_{\text{small}}^2} < 2)$	$((\bar{X}_1 - \bar{X}_2) \pm E)$ $E = t_{\alpha/2, df} \sqrt{\frac{S_p^2}{n_1} + \frac{S_p^2}{n_2}}$ • pooled variance (pooled variance) $S_p^2 = \frac{(n_1 - 1) \cdot s_1^2 + (n_2 - 1) \cdot s_2^2}{n_1 + n_2 - 2}$ • $df = n_1 + n_2 - 2$	step 1, 2, 3 (same table chap 9) step 4: Test statistic $t_0 = \frac{\bar{X}_1 - \bar{X}_2 - \Delta_0}{\sqrt{\frac{S_p^2}{n_1} + \frac{S_p^2}{n_2}}}$
ii) not assume $\sigma_1^2 = \sigma_2^2$ $(\frac{s_{\text{large}}^2}{s_{\text{small}}^2} > 2)$	$((\bar{X}_1 - \bar{X}_2) \pm E)$ $E = t_{\alpha/2, df} \sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}$ $df = \frac{(\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2})}{\frac{(S_1^2/n_1)^2}{n_1 - 1} + \frac{(S_2^2/n_2)^2}{n_2 - 1}}$	step 1, 2, 3 (same table chap 9) step 4: Test statistic $t_0 = \frac{\bar{X}_1 - \bar{X}_2 - \Delta_0}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}}$

② For difference ~~mean~~ proportion ($p_1 - p_2$).

Point estimate for $p_1 - p_2$ $\hat{p}_1 - \hat{p}_2$	CI for $p_1 - p_2$ $((\hat{p}_1 - \hat{p}_2) \pm E)$	Test $H_0: p_1 - p_2 = 0$
$\hat{p}_1 = \frac{x_1}{n_1}$ $\hat{p}_2 = \frac{x_2}{n_2}$	$E = z_{\alpha/2} \sqrt{\frac{\bar{p}(1-\bar{p})}{n_1} + \frac{\bar{p}(1-\bar{p})}{n_2}}$	step 1, 2, 3 (same chap 9) step 4: $z_0 = \frac{\hat{p}_1 - \hat{p}_2 - 0}{\sqrt{\bar{p}(1-\bar{p}) (\frac{1}{n_1} + \frac{1}{n_2})}}$
$\bar{p}$: pooled proportion $\bar{p} = \frac{x_1 + x_2}{n_1 + n_2}$		

CHAPTER 11:

Solution.

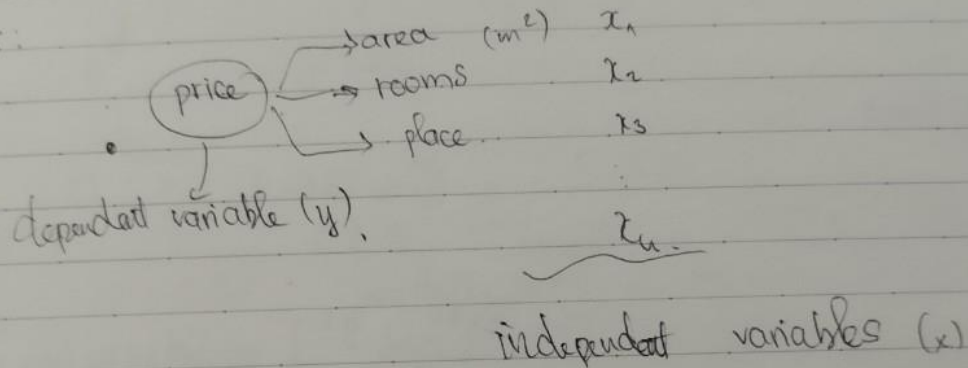
$$t_{0.05,9} = 2.685$$

$$E = t_{0.05,9} \cdot \frac{s}{\sqrt{n}} = 2.685 \cdot \frac{1.83}{\sqrt{10}} = 1.55$$

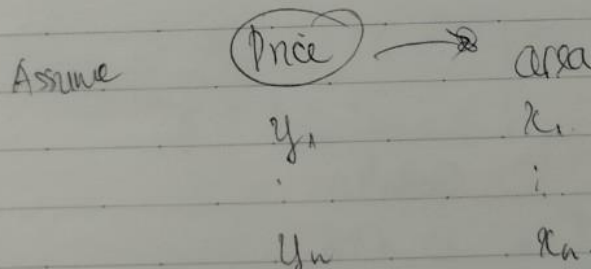
$$(9: (3.1, 6.3))$$

CHAP 11: Simple Linear regression and correlation

Ex:



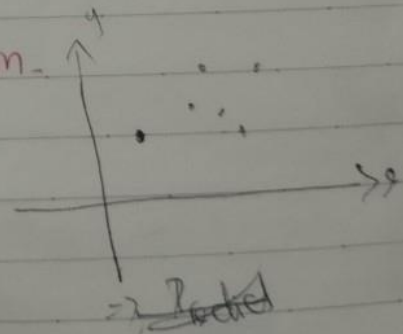
→ Multiple linear regression.



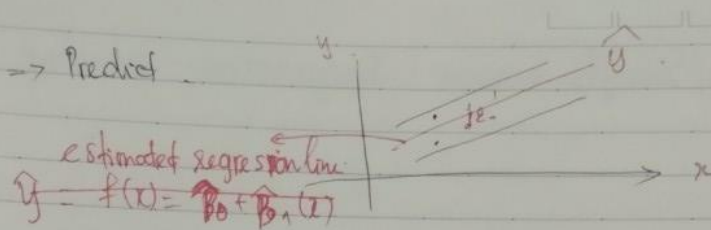
Simple linear regression.

pair data (x_i, y_i) .

→ Scatter plot



→ Predict

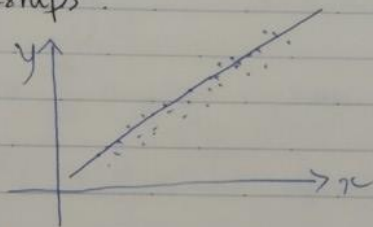


$$\hat{y} = f(x) = \hat{\beta}_0 + \hat{\beta}_1(x)$$

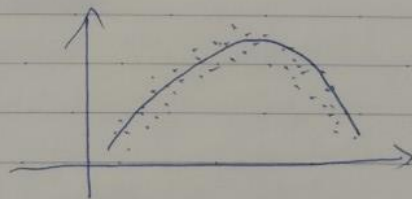
$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$$

$$\varepsilon_i = y_i - \hat{y}_i \quad \begin{cases} \text{residual error} \\ \text{random error} \end{cases}$$

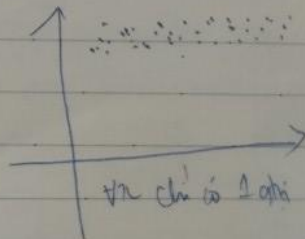
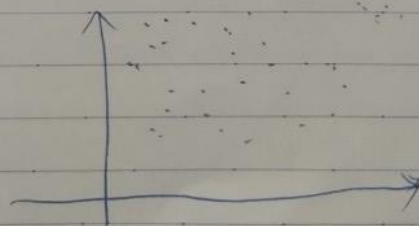
Linear relationships:



Curved linear relationships:



No relationships:

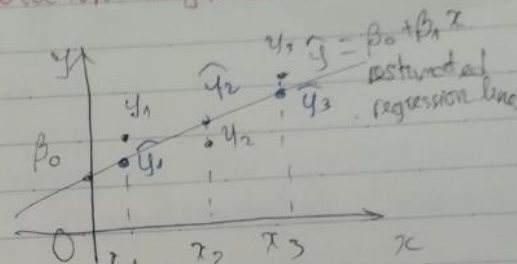


⊛ Simple Linear Regression Model.

$$y = \underbrace{\beta_0 + \beta_1 x}_{\hat{y} = f(x)} + \underbrace{\varepsilon}_{\text{random error}}$$

β_0 - y-intercept of population regression line
 β_1 - slope of population regression line

$\varepsilon_i = y_i - \hat{y}_i$
 ↙ observed data at $x = x_i$
 ↘ predicted value at $x = x_i$



sum square of error. $SS_E = \sum_{i=1}^n \varepsilon_i^2 \Rightarrow \min$

Assume $\varepsilon \sim N(0, \sigma^2)$.
 estimated y-intercept $\hat{\beta}_0 \sim \beta_0$
 estimated slope $\hat{\beta}_1 \sim \beta_1$
 $\hat{\sigma}^2 \sim \sigma^2$
 estimated of standard error.

$$L = \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2$$

$$\Rightarrow L = \|Y - \theta X\|^2$$

$$\text{Min } L: \frac{dL}{d\theta} = 0$$

$$\theta = \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix}$$

$$\Rightarrow \theta = (X^T X)^{-1} X^T Y$$

$$SS_E = \sum_{i=1}^n \varepsilon_i^2 = \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2 \rightarrow \min$$

$$\Rightarrow \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}, \quad \bar{y} = \frac{\sum y_i}{n}, \quad \bar{x} = \frac{\sum x_i}{n}$$

$$\hat{\beta}_1 = \frac{\sum x_i y_i - \frac{\sum x_i \sum y_i}{n}}{\sum x_i^2 - \frac{(\sum x_i)^2}{n}}$$

Def: The fitted or estimated regression line for data (x_i, y_i) , $i = 1, \dots, n$

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$$

$$S_{xy} = \sum (x_i - \bar{x})(y_i - \bar{y}) = \sum x_i y_i - n \bar{x} \bar{y}$$

$$S_{xx} = \sum (x_i - \bar{x})^2 = \sum x_i^2 - n \bar{x}^2$$

$$S_{yy} = \sum (y_i - \bar{y})^2 = \sum y_i^2 - n \bar{y}^2$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}}$$

$y_2 - \hat{y}_1 = \hat{\beta}_1$: estimates change is mean value of y when x increase 1 unit

+ If product ↑ then the value of cost ↑ average

- Sample coefficient of: $R = \sqrt{\frac{SSR}{SST}}$

$$\hat{\sigma} = \sqrt{\frac{SSE}{n-2}}$$

* Standard error of estimate

total variance (total sum of square)

$$SST = \sum (y_i - \bar{y})^2 = S_{yy}$$

Error sum of square

$$SSE = \sum (y_i - \hat{y}_i)^2 \leftarrow \text{unexplained variation}$$

• Regression sum of square:

$$SSR = \sum (\hat{y}_i - \bar{y})^2 \leftarrow \text{explained variation.}$$

$$SSE + SSR = SST$$

* Unbiased estimator of standard error of estimate

$$\text{is } \hat{\sigma} = \sqrt{\frac{SSE}{n-2}}$$

* Coefficient of determination: $R^2 = \frac{SSR}{SST}$

↳ variation in Y is explained by variation in X
 $0.99 \Rightarrow$ correlation: strong.

†) $0 \leq R^2 \leq 1$

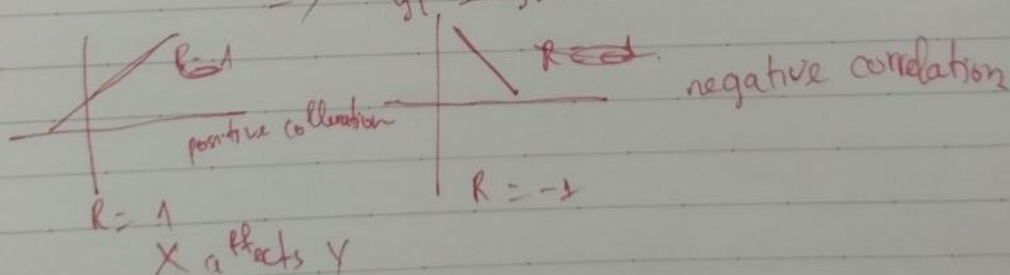
$$R^2 = 1 \quad SSR = SST$$

$$\Rightarrow SSR = SST$$

$$SSE = 0$$

$$\Rightarrow \sum (y_i - \hat{y}_i)^2 = 0$$

$$\Rightarrow y_i = \hat{y}_i$$

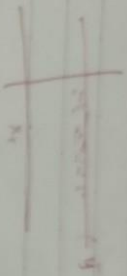


$$R^2 = 0$$

$$\Rightarrow SSR = 0$$

$$\Rightarrow \sum (\hat{y}_i - \bar{y})^2 = 0$$

$$\Rightarrow \hat{y}_i = \bar{y}$$

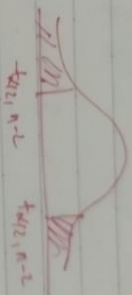


no relationship
no correlation
Use $\hat{y}$ to predict the value of y
at $x = x_1$

* Test of hypotheses about the slope β_1

$$H_0 = \beta_1 = \beta_{1,0} \quad (1\text{st. null test})$$

$$H_1 = \beta_1 \neq \beta_{1,0}$$



* Find critical value: $\pm t_{n/2, n-2}$

* Find test statistic: $t_0 = \frac{\hat{\beta}_1 - \beta_{1,0}}{\text{se}(\hat{\beta}_1)}$

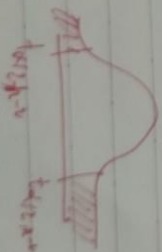
estimated standard error of slope

$$\text{se}(\hat{\beta}_1) = \sqrt{\frac{s^2}{S_x}}$$

* Test hypothesis for y -intercept β_0

$$H_0: \beta_0 = \beta_{0,0}$$

$$H_1: \beta_0 \neq \beta_{0,0}$$



$$t_0 = \frac{\hat{\beta}_0 - \beta_{0,0}}{\text{se}(\hat{\beta}_0)} \rightarrow \text{estimated standard error of } y\text{-intercept}$$

$$\text{se}(\hat{\beta}_0) = \sqrt{s^2 \left(\frac{1}{n} + \frac{x_1^2}{S_x} \right)}$$

* Coefficient of correlation (sample)

$$R = \sqrt{r^2} = \sqrt{\frac{SSR}{SST}}$$

$$R = \frac{s_{xy}}{\sqrt{s_{xx} \cdot s_{yy}}}$$

$$s_{xx} = \sum (x_i - \bar{x})^2 = \sum x_i^2 - n\bar{x}^2 \quad (= (n-1)s_x^2)$$

$$s_{yy} = \sum (y_i - \bar{y})^2 = \sum y_i^2 - n\bar{y}^2 \quad (= (n-1)s_y^2)$$

$$s_{xy} = \sum (x_i - \bar{x})(y_i - \bar{y}) = \sum x_i y_i - n\bar{x}\bar{y}$$

$$\bar{x} = \frac{\sum x_i}{n}, \quad \bar{y} = \frac{\sum y_i}{n}$$

$$0 \leq R^2 \leq 1$$

$$-1 \leq R \leq 1$$

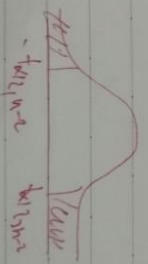
• R and $\hat{\beta}_1$: same sign

$R = \hat{\rho}$: unbiased estimator

(population coefficient of correlation)

$$H_0: \rho = 0$$

$$H_1: \rho \neq 0$$



$$t_0 = \frac{R - 0}{\sqrt{\frac{1 - R^2}{n-2}}}$$